Joseph Catanzarite

AI Researcher | Data Scientist | Machine Learning, LLMs & Astrophysics

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jcatanza.github.io | Google Scholar

PROFESSIONAL SUMMARY

AI researcher and data scientist with extensive experience applying machine learning, Bayesian statistics, and scientific computing to NASA missions and teaching. Built a Random Forest classifier (97% accuracy) for Kepler exoplanet candidate vetting, published in NASA's DR24 catalog; led JPL's analysis team in a five-team double-blind exoplanet detection test. Completing an M.S. in Artificial Intelligence at Johns Hopkins (expected 2028) with hands-on work in large language models (LLMs), generative AI, and AI agents. Author of peer-reviewed astrophysics papers (~5,900 citations, h-index 32). Instructor and research mentor in AI, data science, physics, and astronomy.

CORE SKILLS

AI & Machine Learning: Large Language Models (LLMs), Generative AI, AI Agents, Deep Learning, Neural Networks (CNN, VAE, GAN), Natural Language Processing (NLP), Computer Vision, Supervised Learning, Random Forest, k-Nearest Neighbors, Model Training & Evaluation, LLMOps, Claude (Anthropic)

Statistics & Modeling: Bayesian Modeling & Inference, Statistical Inference, Hierarchical Modeling, Monte Carlo Simulation, Time Series Analysis, Signal Processing, Wavelet Analysis, Experimental Design, Causal Inference, Optimization

Scientific Computing & Astrophysics: Exoplanet Detection, Astrometry, Radial Velocity, Transit Photometry, Astrophysical Data Analysis, Numerical Simulation, Data Validation, Pipeline Development, High-Performance Computing

Programming & Tools: Python, MATLAB, SQL, PyTorch, TensorFlow, Keras, fastai, scikit-learn, NumPy, pandas, Jupyter, Git/GitHub, SVN, Google Cloud Platform

Teaching & Leadership: Curriculum Development, Teaching & Mentoring, Research Mentorship, Technical Writing

PROFESSIONAL EXPERIENCE

AI and Data Science Mentor (Learning Coach) | Elton University / *Contract, Remote* *Oct 2026 – Present*

- Supervise and guide hands-on student projects and education in artificial intelligence, machine learning, and data science.

LLM Researcher, Handshake AI Fellow | Handshake AI / *Freelance, Remote* *Sep 2026 – Present*

- Conduct research and evaluation work on AI agents and large language models (LLMs).

Research Mentor in AI, Astronomy, and Physics | Nova Scholar Education / *Part-time, Remote* *Apr 2026 – Present*

- Mentor young researchers one-on-one on astrophysics and machine learning projects that lead to journal publications, developing their research, critical thinking, and scientific communication skills.

Career Break | Family and well-being; preparation for AI graduate school *Jan 2022 – Dec 2025*

Data Scientist / Data Science Instructor | BloomTech / *Full-time, Remote* *Aug 2021 – Dec 2021*

- Developed and taught a Data Science and Machine Learning curriculum for an online Bachelor's degree program; supervised student projects and work-study students.

Data Scientist / Lead Data Science Instructor | Make School / *Full-time, Remote* *Nov 2020 – Aug 2021*

- Developed and delivered Deep Learning & Computer Vision, Machine Learning in Production, and Intro to Data Science courses; led the capstone Data Science Technical Seminar (research to deployment).

Volunteer & Fellowship Roles in AI | DataKind, TWiML & AI, University of San Francisco *Oct 2018 – Apr 2021*

- **Volunteer Data Scientist, DataKind Medic Mobile (2019–2020):** improved models for maternal and child health outcomes.
- **Volunteer AI Study Group Lead, TWiML & AI (2019–2021):** led remote study groups on machine learning, NLP (CS224N), and deep learning math.
- **Fast.ai International Fellow, USF (2018–2020):** organized study groups and developed fast.ai deep learning and NLP curriculum.

Data Scientist – NASA Kepler Mission Science Office | SETI Institute / *Mountain View, CA* *Jan 2013 – Aug 2017*

- As astronomer and scientific programmer, built a Random Forest "autovetter" that classifies Kepler exoplanet candidates with 97% accuracy; published in NASA's Kepler DR24 catalog and documented in the Autovetter Planet Candidate Catalog report.
- Led synthetic flux-level transit injection experiments for Kepler's Occurrence Rate Data Products (delivered to NASA Headquarters); applied k-Nearest Neighbors (kNN) to estimate detection efficiency versus signal-to-noise ratio, cutting NASA Pleiades supercomputer time by 85%.
- Created Kepler Data Processing Pipeline algorithms in MATLAB under SVN: wavelet-based denoiser (Presearch Data Conditioning), ghost diagnostic and ephemeris matching (Data Validation), thruster firing detector (Photometric Analysis).

Project Staff Scientist / Research Technologist IV | NASA Jet Propulsion Laboratory (JPL) / *Pasadena, CA* *Jan 2003 – Jan 2013*

- As an astronomer on the SIM planet-finding team, led JPL's Analysis Team in NASA's double-blind test for the Space Interferometry Mission (SIM), coordinating five teams to reach consensus solutions for planet masses and orbits from joint astrometric and radial velocity data.
- Invented a novel hierarchical algorithm (nonlinear orbit fitting) to detect exoplanets and characterize their masses and orbits; played a key role in demonstrating SIM's capacity to detect Earth-mass planets.
- Ran mission-level Monte Carlo simulations of exoplanet-finding capability (astrometric, radial velocity, coronagraphic) and starspot effects; devised wavefront-sensing and line-of-sight jitter algorithms for the Advanced Mirror Development Testbed.

Earlier Experience

- **Senior Engineer and Senior Application Developer, California Institute of Technology:** trade studies of astrometric performance; data analysis algorithms for SIM's Micro-Arcsecond Metrology testbed; WIRE star-tracker data preprocessing for asteroseismology research.
- **Professor of Physics and Astronomy, Cypress College:** chaired the Physical Sciences Department; developed and taught physics and astronomy courses (lecture and laboratory).

SELECTED AI PROJECTS

Professor.Claude.AI — autonomous AI research agent | github.com/jcatanza/professor-claud-ai *May 2026 – Present*

- Python/LLMOps pipeline that nightly ingests new arXiv papers, triages them, and deep-reads the best with Claude into structured analyses; persistent semantic memory; daily email and GitHub digest.

Chinese Character Recognition and Generation — Johns Hopkins *2026*

- Compared Conditional VAE, Conditional GAN, and hybrid C-VAEGAN generative data augmentation on Chinese-MNIST (FID, SSIM, LPIPS, downstream CNN accuracy); designed a novel two-stage quality filter for synthetic samples.

Shots, Not Stars! 2026 FIFA World Cup Prediction Model | jcatanza.github.io/FIFA-World-Cup-2026-Predictions *Jul 2026*

- Random Forest on 17 team-level statistics from 67 matches with bootstrap confidence estimates; 62% win probability for Spain closely matched Polymarket; shots and saves were the top predictors.

A Primer on Modal and Epistemic Logic | jcatanza.github.io/logic *2026*

- Self-contained tutorial from propositional logic to modal and epistemic proofs about Belief-Desire-Intention (BDI) agents.

Science & Machine Intelligence — LinkedIn newsletter | Subscribe on LinkedIn *Monthly*

- Monthly newsletter (203 subscribers) on current boundaries and future trends in AI and scientific discovery.

PUBLICATIONS

~5,900 citations | h-index 32 | i10-index 52 — full list on Google Scholar

- **First author:** "Catching Up with the Fastest Star in the Galaxy: A Two-Passage GRAVITY+ Campaign to Detect S301's Schwarzschild Precession," arXiv:2609.06936 (2026), submitted to PASP; Catanzarite & Shao, "The Occurrence Rate of Earth Analog Planets Orbiting Sun-like Stars," ApJ 738, 151 (2011); "Astrometric Detection of Terrestrial Planets in the Habitable Zones of Nearby Stars with SIM PlanetQuest," PASP 118, 1319 (2006).
- **Kepler team (selected):** PASP 126 (2014), wavelet error correction; ApJ 809 (2015), planet occurrence rates; AJ 150 (2015), Kepler-452b; ApJS 224 (2016), planet candidate catalog; PASP 130 (2018), Data Validation; AJ 161 (2021), rocky habitable-zone planets.

EDUCATION

M.S., Artificial Intelligence | Johns Hopkins University, Whiting School of Engineering *Jan 2026 – May 2028 (expected)*
M.S., Electrical Engineering | University of Southern California (Rockwell Fellowship)
M.S., Physics | University of Arizona (Mathematics minor)
B.S., Mathematics | California Institute of Technology (Caltech)

CERTIFICATIONS & TRAINING

Deep learning & ML: Practical Deep Learning for Coders 1 & 2 (fast.ai/USF, 2018); Deep Learning Specialization (deeplearning.ai, 2018); Machine Learning Specialization (U. Washington, 2017); Foundations of Statistical ML (ASU, 2018); TensorFlow 2 (Imperial College London, 2021); Machine Learning on Google Cloud (2018).
Statistics, math & data: Learning from Data (Caltech); Convex Optimization (Stanford); Probability (MIT/edX); Causal Inference (Penn); SQL (UC Davis); Python for Research (HarvardX).

HONORS & AWARDS

- NASA JPL Group Achievement Awards (5): SIM Science Office Team (2011); SIM Planet Finding Capability Study (2010); SIM Planet Hunter Mission Study (2007); SIM Microarcsecond Metrology Testbed (2003, 2005)
- NASA JOVE Research Fellowship; NASA Summer Faculty Fellowship; USC Rockwell Fellowship; NISOD Excellence Award; Cypress College Faculty Development Award